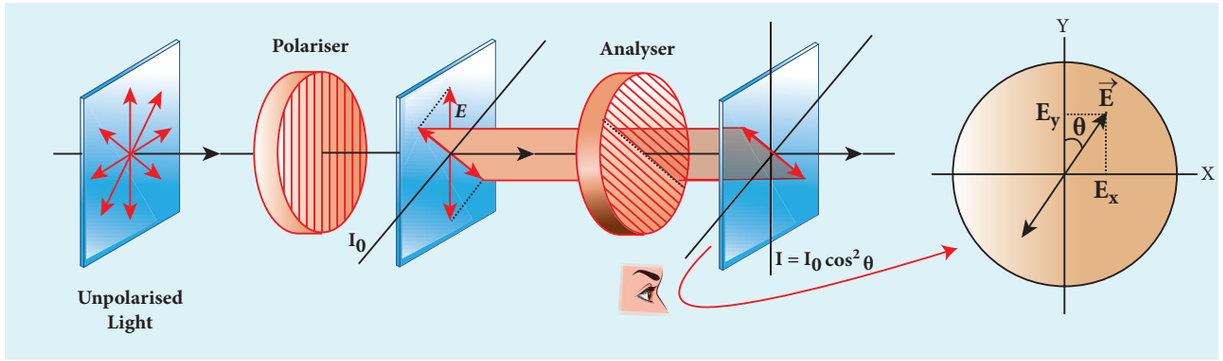


Figure 7.29 Malus' law

on an analyser, the intensity of light I transmitted from the analyser varies directly as the square of the cosine of the angle θ between the transmission axes of polariser and analyser as shown in Figure 7.29. This is known as Malus' law.

$$I = I_0 \cos^2 \theta \qquad (7.55)$$

The proof of Malus' law is as follows. Let us consider that the transmission axes of the polariser and the analyser are inclined by an angle θ is as shown in Figure 7.30. Let I_0 be the intensity and a be the amplitude of the electric vector transmitted by the polariser. The amplitude a of the incident light has two rectangular components, $a\cos\theta$ and $a\sin\theta$ which are the parallel and perpendicular components to the axis of transmission of the analyser.

Only the component $a\cos\theta$ will be transmitted by the analyser. The intensity of light transmitted from the analyser is proportional to the square of the component of the amplitude transmitted by the analyser.

$$I \propto (a\cos\theta)^2$$

$$I = k(a\cos\theta)^2$$

Where k is constant of proportionality.

$$I = ka^2 \cos^2 \theta$$

$$I = I_0 \cos^2 \theta$$

Where, $I_0 = ka^2$ is the maximum intensity of light transmitted through the analyser.

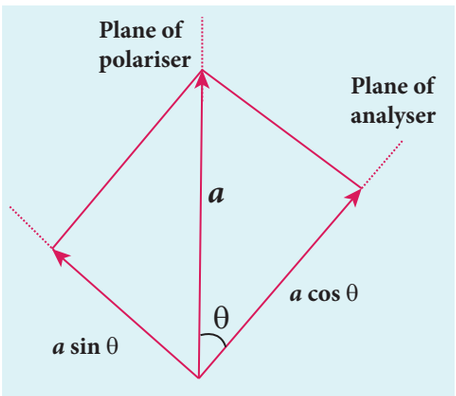


Figure 7.30 Malus' law

The following are few special cases.

Case (i) When $\theta = 0^o$, $\cos 0^o = 1$, $I = I_0$

When the transmission axis of polariser is parallel to that of the analyser, the intensity of light transmitted from the analyser is equal to the incident light that falls on it from the polariser.

Case (ii) When $\theta = 90^o$, $\cos 90^o = 0$, $I = 0$

When the transmission axes of polariser and analyser are perpendicular to each other, the intensity of light transmitted from the analyser is zero.

UNIT-7(XII-Physics_Vol-2)_new.indd 82 21-07-2020 16:36:17